

RAHUL KUMAR

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Education

BMS College of Engineering, Bangalore

November 2022 - June 2026

B.Tech : Electronics and Communication Engineering

CGPA: 7.5/10.0

Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming, Database Management System, Wireless Communication Networks, Computer Networks, Web Development, VLSI, Digital Communications

Projects

Web : Expense Tracker Application | React, Node.js, MongoDB, REST API, JWT | April 2025 – July 2024

- Built a **full-stack expense tracker** using **MERN Stack** for managing and visualizing daily expenses.
- Designed responsive React UI with **Login, Signup, and Dashboard screens**.
- Implemented **JWT** authentication, **CRUD APIs**, and **MongoDB** for secure data storage.

Web : Sorting Visualizer | HTML, CSS, JavaScript, Font Awesome, Google Fonts | July 2025 – October 2025

- Developed a full-stack web application for visualizing five sorting algorithms (**Bubble, Quick, Insertion, Selection, Merge**) with interactive, **real-time animations and graph views (Bar Graph)**, handling datasets up to 100 elements and improving user engagement by 40%.
- Implemented dynamic array controls and animation speed for real-time visualizations. Enabling features such as start, pause, and resume for enhanced user control during live data visualization

Embedded : Surveillance Robot | ESP32-CAM, Arduino IDE, HTML, CSS, L298N | October 2024 – December

- Built a **Wi-Fi-controlled surveillance robot** using **ESP32-CAM** for **real-time video streaming** and remote control.
- Configured ESP32-CAM to create its own Wi-Fi access point, accessible via IP address in any browser.
- Developed a custom web interface using **HTML, CSS, and JavaScript** to display live camera feed and control buttons.
- Programmed motor movement (forward, backward, left, right) using **L298N motor driver** through web commands.
- Coded and uploaded firmware in **Arduino IDE**, integrating both camera and motor control features seamlessly.

Technical Skills

Languages: C/C++, Java, HTML, SQL, Power BI, Python, CSS

Framework:: React, Node.js, Mongoose, CSS

Tools/ Technologies: Git, GitHub, VS Code,

Databases: MySQL, MongoDB

Soft Skills: Problem-solving, Collaboration, Adaptability

Certifications and Achievements

- **DSA Enthusiast** – Solved **190+** problems on LeetCode and GFG; honed problem-solving and data structure mastery.
- **Hackathon Participant** – Built real-world solutions in collaborative coding events.
- **Rotaract Club Member** – Engaged in leadership, teamwork, and community service projects fostering social responsibility.